STAT 440/540 Milestone 2: EDA & Initial Models

Group 03

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1 Overview

- Briefly restate question, dataset, and key variables.

The purpose of the this project is to explore the relationship that mental health and other life style factors that can affect a college student's GPA. This dataset gathers data on GPA, mental health factors, student habits, and other life style factors. The key variables in this data set is the repsonse variable *GPA* and other important variables such as sleep and [something else].

1.1 Package Setup

```r
if (!requireNamespace("pacman", quietly = TRUE)) install.packages("pacman")
pacman::p_load(tidyverse, janitor, skimr, GGally, broom, patchwork, knitr, kableExtra)
set.seed(440540)
```

1.2 Data Import and Cleaning

```r
raw <- readr::read_csv("../data/raw/student_lifestyle_dataset.csv") |> janitor::clean_names()

# raw <- readr::read_csv("../data/raw/example_stub.csv") |> janitor::clean_names()

summary(raw)
```

```
   student_id     study_hours_per_day extracurricular_hours_per_day
 Min.   :   1.0   Min.   : 5.000      Min.   :0.00
 1st Qu.: 500.8   1st Qu.: 6.300      1st Qu.:1.00
 Median :1000.5   Median : 7.400      Median :2.00
 Mean   :1000.5   Mean   : 7.476      Mean   :1.99
 3rd Qu.:1500.2   3rd Qu.: 8.700      3rd Qu.:3.00
 Max.   :2000.0   Max.   :10.000      Max.   :4.00
 sleep_hours_per_day social_hours_per_day physical_activity_hours_per_day
 Min.   : 5.000      Min.   :0.000        Min.   : 0.000
 1st Qu.: 6.200      1st Qu.:1.200        1st Qu.: 2.400
 Median : 7.500      Median :2.600        Median : 4.100
 Mean   : 7.501      Mean   :2.705        Mean   : 4.328
 3rd Qu.: 8.800      3rd Qu.:4.100        3rd Qu.: 6.100
 Max.   :10.000      Max.   :6.000        Max.   :13.000
      gpa        stress_level
 Min.   :2.240   Length:2000
 1st Qu.:2.900   Class :character
 Median :3.110   Mode  :character
 Mean   :3.116
 3rd Qu.:3.330
 Max.   :4.000
```

```r
# dat <- raw

dat <- raw |>
  dplyr::select(-student_id)



# Code originally given
  # mutate(
  #   x3 = factor(x3)
  # ) |>
  # tidyr::drop_na(y)
```

1.3 Descriptive Statistics

```r
##| tbl-cap: "Summary statistics for numeric variables."

summary_data <- dat |>
  dplyr::select(where(is.numeric)) |>
  skimr::skim()

# First half of kable
summary_data |>
  dplyr::select(
    skim_type,
    skim_variable,
    n_missing,
    complete_rate,
    numeric.mean,
    numeric.sd,
  ) |>
  knitr::kable()
# Second half of kable
summary_data |>
  dplyr::select(
    skim_variable,
    numeric.p0,
    numeric.p25,
    numeric.p50,
    numeric.p75,
    numeric.p100
  ) |>
  knitr::kable()
# THE original CODE GIVEN
# dat |>
#   dplyr::select(where(is.numeric)) |>
#   skimr::skim() |>
#   knitr::kable()
```

Table 1

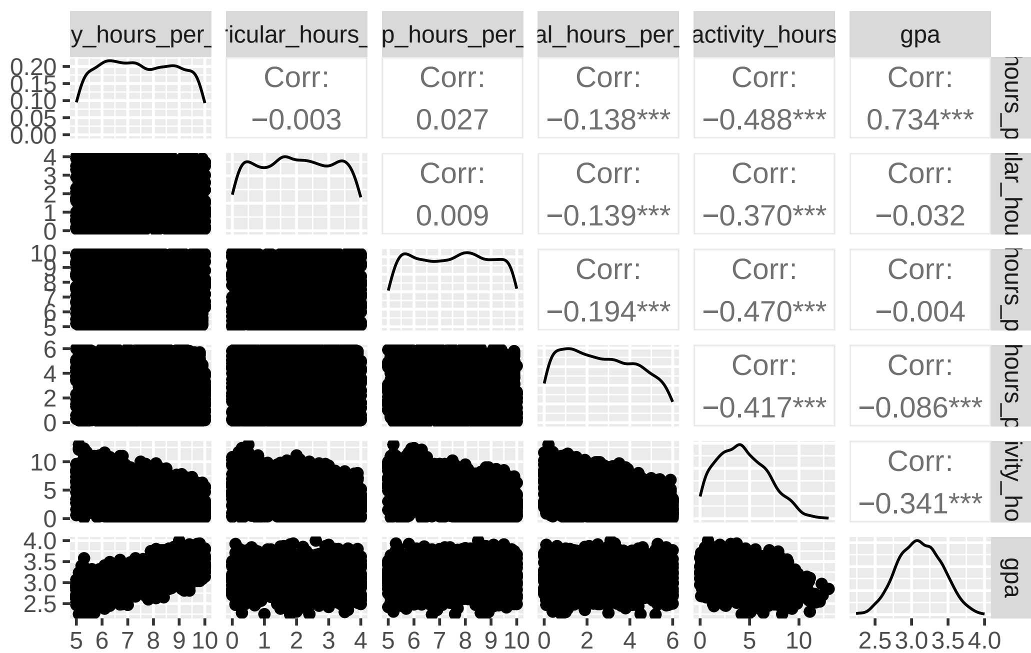
skim_type	skim_variable	n_missing	complete_rate	numeric.mean	numeric.sd
numeric	study_hours_per_day	0	1	7.47580	1.4238884
numeric	extracurricular_hours_per_day	0	1	1.99010	1.1558547
numeric	sleep_hours_per_day	0	1	7.50125	1.4609485
numeric	social_hours_per_day	0	1	2.70455	1.6885141
numeric	physical_activity_hours_per_day	0	1	4.32830	2.5141101
numeric	gpa	0	1	3.11596	0.2986735

Table 2

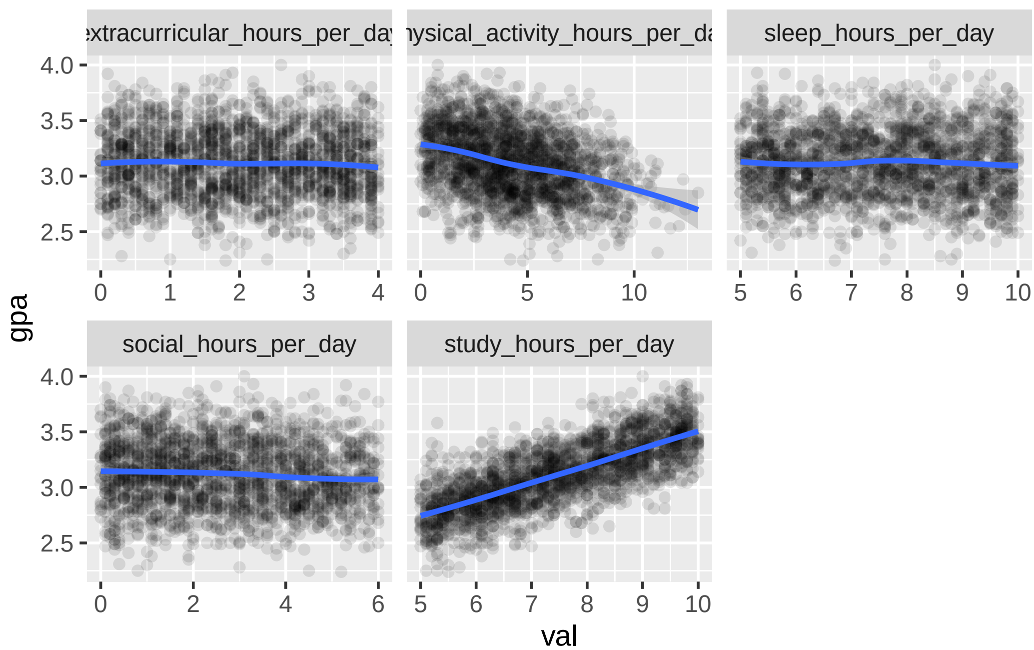
skim_variable	numeric.p0	numeric.p25	numeric.p50	numeric.p75	numeric.p100
study_hours_per_day	5.00	6.3	7.40	8.70	10
extracurricular_hours_per_day	0.00	1.0	2.00	3.00	4
sleep_hours_per_day	5.00	6.2	7.50	8.80	10
social_hours_per_day	0.00	1.2	2.60	4.10	6
physical_activity_hours_per_day	0.00	2.4	4.10	6.10	13
gpa	2.24	2.9	3.11	3.33	4

1.4 Exploratory Visualizations

```r
##| label:fig-pairs
##| fig-cap: "Pairs plot for numeric variables."

# ---- From old data set
# gpa_dat <- dat |>
#   dplyr::select(academic_performance_gpa, study_hours_per_week, financial_stress)

# gpa_dat2 <- dat |>
#   dplyr::select(academic_performance_gpa, mental_stress_level, diet_quality, cognitive_disto

# ----

GGally::ggpairs(dplyr::select(dat, where(is.numeric)))
```



```r
##| label:fig-y-vs-x
##| fig-cap: "Outcome vs. selected predictors with loess smooth."
dat |>
  dplyr::select(where(is.numeric)) |>
  tidyr::pivot_longer(cols = -c(gpa), names_to = "var", values_to = "val") |>
  ggplot(aes(val, gpa)) +
  geom_point(alpha = 0.1) +
  geom_smooth(method = "loess", se = TRUE) +
  facet_wrap(~ var, scales = "free_x")
```




```r
  # dat |>
  # dplyr::select(where(is.numeric)) |>
  # tidyr::pivot_longer(cols = -c(y), names_to = "var", values_to = "val") |>
  # ggplot(aes(val, y)) +
  # geom_point(alpha = 0.5) +
  # geom_smooth(method = "loess", se = TRUE) +
  # facet_wrap(~ var, scales = "free_x")
```

1.5 Initial Simple Regressions

```r
## | label: simple-models
cand_x <- c("study_hours_per_day","extracurricular_hours_per_day","sleep_hours_per_day")
fits <- lapply(cand_x, \(v) lm(reformulate(v, response = "gpa"), data = dat))
models <- tibble::tibble(
  predictor = cand_x,
```

```
  glance = purrr::map(fits, broom::glance),
  tidy = purrr::map(fits, broom::tidy)
)

dplyr::bind_rows(purrr::map2(models$predictor, models$glance, \(v,g) dplyr::mutate(g, predicto
  dplyr::select(predictor, r.squared, adj.r.squared, AIC, BIC, p.value) |>
  knitr::kable(digits = 3, caption = "Simple regression comparisons")
```

Table 3: Simple regression comparisons

predictor	r.squared	adj.r.squared	AIC	BIC	p.value
study_hours_per_day	0.539	0.539	-703.501	-686.698	0.000
extracurricular_hours_per_day	0.001	0.001	845.066	861.869	0.150
sleep_hours_per_day	0.000	0.000	847.101	863.903	0.848

```
# cand_x <- c("x1","x2","x3")
# fits <- lapply(cand_x, \(v) lm(reformulate(v, response = "y"), data = dat))
# models <- tibble::tibble(
#   predictor = cand_x,
#   glance = purrr::map(fits, broom::glance),
#   tidy = purrr::map(fits, broom::tidy)
# )

# dplyr::bind_rows(purrr::map2(models$predictor, models$glance, \(v,g) dplyr::mutate(g, predic
#   dplyr::select(predictor, r.squared, adj.r.squared, AIC, BIC, p.value) |>
#   knitr::kable(digits = 3, caption = "Simple regression comparisons")
```

1.6 Early Findings and Next Steps

- What looks promising? Any evidence of nonlinearity or interactions to explore?

1.7 Reproducibility Snapshot

```
##| label: session-info
sessionInfo()
```

```
R version 4.5.0 (2025-04-11)
Platform: aarch64-apple-darwin20
Running under: macOS Sonoma 14.3

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
```

```
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  LA

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/Denver
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
 [1] kableExtra_1.4.0 knitr_1.50       patchwork_1.3.0  broom_1.0.8
 [5] GGally_2.2.1     skimr_2.2.1      janitor_2.2.1    lubridate_1.9.4
 [9] forcats_1.0.0    stringr_1.5.1    dplyr_1.1.4      purrr_1.0.4
[13] readr_2.1.5      tidyr_1.3.1      tibble_3.2.1     ggplot2_3.5.2
[17] tidyverse_2.0.0

loaded via a namespace (and not attached):
 [1] gtable_0.3.6       xfun_0.52          lattice_0.22-6     tzdb_0.5.0
 [5] vctrs_0.6.5        tools_4.5.0        generics_0.1.4     parallel_4.5.0
 [9] pacman_0.5.1       pkgconfig_2.0.3    Matrix_1.7-3       RColorBrewer_1.1-3
[13] lifecycle_1.0.4    compiler_4.5.0     farver_2.1.2       textshaping_1.0.1
[17] repr_1.1.7         snakecase_0.11.1   htmltools_0.5.8.1  yaml_2.3.10
[21] pillar_1.10.2      crayon_1.5.3       nlme_3.1-168       ggstats_0.9.0
[25] tidyselect_1.2.1   digest_0.6.37      stringi_1.8.7      labeling_0.4.3
[29] splines_4.5.0      fastmap_1.2.0      grid_4.5.0         cli_3.6.5
[33] magrittr_2.0.3     base64enc_0.1-3    withr_3.0.2        scales_1.4.0
[37] backports_1.5.0    bit64_4.6.0-1      timechange_0.3.0   rmarkdown_2.29
[41] bit_4.6.0          hms_1.1.3          evaluate_1.0.3     viridisLite_0.4.2
[45] mgcv_1.9-1         rlang_1.1.6        Rcpp_1.0.14        glue_1.8.0
[49] xml2_1.3.8         svglite_2.2.1      rstudioapi_0.17.1  vroom_1.6.5
[53] jsonlite_2.0.0     R6_2.6.1           plyr_1.8.9         systemfonts_1.2.3
```